Lecture 6. Nonparametric Estimation of Basic Quantities for Right-Censored Data (Part 2)

Pointwise Confidence Intervals and Confidence Bands

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1 Recap

$$\hat{S}(t) = \prod_{t_j \le t} \Big(1 - \frac{d_j}{Y_j}\Big), \quad \hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)}; \qquad \tilde{H}(t) = \sum_{t_j \le t} \frac{d_j}{Y_j}, \quad \sigma_H^2(t) = \sum_{t_j \le t} \frac{d_j}{Y_j^2}.$$

Throughout, define

$$\sigma_S^2(t) = \frac{\hat{V}[\hat{S}(t)]}{\hat{S}^2(t)} = \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)},$$

so that $\hat{S}(t)\sigma_S(t)$ is the Greenwood standard error.

2 Pointwise Confidence Intervals for $S(t)$

Motivation. An estimate of $S(t)$ needs an interval. Greenwood's variance gives one directly, but the interval may leave $[0, 1]$; transformations such as log and log-log fix this.

2.1 Linear interval

Definition 6.1: Linear CI

$$\Big[\hat{S}(t) - Z_{1-\alpha/2}\hat{S}(t)\sigma_S(t),\quad \hat{S}(t) + Z_{1-\alpha/2}\hat{S}(t)\sigma_S(t)\Big].$$

It is symmetric about $\hat{S}(t)$ and can fall outside $[0, 1]$.

2.2 Transformed intervals: the general recipe

Derivation: delta method for a transformation g

Let g be known, monotone and differentiable. By the delta method, $\mathrm{SE}\{g(\hat{S})\} \approx |g'(\hat{S})| \cdot \mathrm{SE}(\hat{S}) = |g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t)$. A $100(1-\alpha)\%$ CI for $g(S(t))$ is

$$g(\hat{S}(t)) \pm Z_{1-\alpha/2}\,|g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t),$$

and re-transforming gives a CI for $S(t)$:

$$g^{-1}\Big\{g(\hat{S}(t)) \pm Z_{1-\alpha/2}\,|g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t)\Big\}.$$

Why transform? Convergence to normality may be faster on the transformed scale, and the re-transformed CI respects $0 \le S(t) \le 1$.

2.3 Log-log transformation

Theorem 6.2: Log-log CI

$$\Big(\hat{S}(t)^{1/\theta},\ \hat{S}(t)^{\theta}\Big), \qquad \theta = \exp\Big\{\frac{Z_{1-\alpha/2}\,\sigma_S(t)}{\log \hat{S}(t)}\Big\}.$$

Derivation [Supplement]. K&M use $\log H(t) = \log\{-\log S(t)\}$, i.e. $g(u) = \log\{-\log u\}$ with $g'(u) = \dfrac{1}{u \log u}$ and $g^{-1}(v) = \exp\{-\exp(v)\}$. Then

$$|g'(\hat{S})|\hat{S}\sigma_S = \frac{\sigma_S}{|\log \hat{S}|},$$

so the CI for $g(S)$ is $\log\{-\log \hat{S}\} \pm a$ with $a = Z_{1-\alpha/2}\sigma_S/|\log \hat{S}|$. Re-transforming,

$$g^{-1}\big(\log\{-\log \hat{S}\} \pm a\big) = \exp\{\log \hat{S} \cdot e^{\pm a}\} = \hat{S}^{\,e^{\pm a}}.$$

Since $\log \hat{S} < 0$, $\theta = e^{-a} < 1$; the endpoints are $\hat{S}^{1/\theta}$ (lower, as $1/\theta > 1$) and $\hat{S}^{\theta}$ (upper). ■

2.4 Arcsine-square-root transformation

Theorem 6.3: Arcsine-square-root CI

$$\text{lower} = \sin^2\Big\{\max\Big[0,\ \arcsin\big(\hat{S}(t)^{1/2}\big) - 0.5Z_{1-\alpha/2}\sigma_S(t)\Big(\frac{\hat{S}(t)}{1-\hat{S}(t)}\Big)^{1/2}\Big]\Big\},$$

$$\text{upper} = \sin^2\Big\{\min\Big[\frac{\pi}{2},\ \arcsin\big(\hat{S}(t)^{1/2}\big) + 0.5 Z_{1-\alpha/2}\sigma_S(t)\Big(\frac{\hat{S}(t)}{1-\hat{S}(t)}\Big)^{1/2}\Big]\Big\}.$$

Derivation [Supplement]. $g(u) = \arcsin(\sqrt{u})$ has $g'(u) = \frac{1}{\sqrt{1-u}} \cdot \frac{1}{2\sqrt{u}}$, so $|g'(\hat{S})|\hat{S}\sigma_S = \frac{\hat{S}\sigma_S}{2\sqrt{\hat{S}(1-\hat{S})}} = 0.5\,\sigma_S\Big(\frac{\hat{S}}{1-\hat{S}}\Big)^{1/2}$. The max / min keep the angle in $[0, \pi/2]$, and $g^{-1}(v) = \sin^2 v$. ■

2.5 Log transformation

Definition 6.4: Log CI (default of R `survfit`)

$g(S) = \log S$, $g'(u) = 1/u$, so $|g'(\hat{S})|\hat{S}\sigma_S = \sigma_S$:

$$\min\Big(\exp\big\{\log\hat{S}(t) \pm Z_{1-\alpha/2}\sigma_S(t)\big\},\ 1\Big).$$

Less recommended than log-log and arcsine-square-root.

Remark 6.5: Comments

- Log-log and arcsine-square-root CIs have better coverage than the linear CI.
- For very large samples the three methods are equivalent.
- The transformed CIs are *not* symmetric about the point estimate; they are appropriate in small samples where the point estimator is biased and its distribution skewed.

Example 6.6: Toy example, $n = 7$ [Supplement]

T_i	1.5	2.5	1.4	6.2	2.8	5.3	4.5
δ_i	1	0	0	1	1	1	0

Ordered: 1.4^+, 1.5, 2.5^+, 2.8, 4.5^+, 5.3, 6.2.

t_j	d_j	Y_j	$\hat{S}$	σ_S	Linear	Log-log	Arcsine
1.5	1	6	0.833	0.183	(0.535, 1.000*)	(0.273, 0.975)	(0.465, 1.000)
2.8	1	4	0.625	0.342	(0.207, 1.000*)	(0.142, 0.893)	(0.213, 0.949)
5.3	1	2	0.313	0.785	(0.000*, 0.794)	(0.013, 0.734)	(0.006, 0.804)
6.2	1	1	0	—	—	—	—

*The raw linear limits (1.131, 1.043, −0.169) fall outside $[0, 1]$; transformed intervals stay inside automatically. Computation at $t = 2.8$: $\sigma_S^2 = \frac{1}{6\cdot5} + \frac{1}{4\cdot3} = 0.1167$, $\theta = \exp\{1.96 \times 0.3416/\ln 0.625\} = 0.241$, so the log-log CI is $(0.625^{1/0.241}, 0.625^{0.241}) = (0.142, 0.893)$.

3 Pointwise Confidence Intervals for $H(t)$

Based on the Nelson–Aalen $\tilde{H}(t)$ and $\sigma_H(t)$:

- **Linear**: $\tilde{H}(t) \pm Z_{1-\alpha/2}\sigma_H(t)$.

- **Log-transformed** (on H): $[\tilde{H}(t)/\phi,\ \phi\tilde{H}(t)],\ \ \phi = \exp\left[\dfrac{Z_{1-\alpha/2}\sigma_H(t)}{\tilde{H}(t)}\right]$ (since $\mathrm{SE}(\log\tilde{H}) \approx \sigma_H/\tilde{H}$).
- **Arcsine-square-root** (on $S = e^{-H}$):

$$\Big[-2\ln\Big\{\sin\Big(\min\Big[\frac{\pi}{2},\ \arcsin\left(e^{-\tilde{H}(t)/2}\right) + 0.5Z_{1-\alpha/2}\sigma_H(t)\{e^{\tilde{H}(t)}-1\}^{-1/2}\Big]\Big)\Big\},$$
$$-2\ln\Big\{\sin\Big(\max\Big[0,\ \arcsin\left(e^{-\tilde{H}(t)/2}\right) - 0.5Z_{1-\alpha/2}\sigma_H(t)\{e^{\tilde{H}(t)}-1\}^{-1/2}\Big]\Big)\Big\}\Big].$$

4 Confidence Bands

4.1 Pointwise intervals vs. bands

For a fixed t_0, a $100(1-\alpha)\%$ CI $[L_{t_0}, U_{t_0}]$ satisfies $P\{L_{t_0} \le S(t_0) \le U_{t_0}\} = 1-\alpha$; once t_0 is fixed, $S(t_0)$ is just a scalar. Hence:

- a plot of pointwise intervals over all t should *not* be read as an interval for the entire survival curve;
- such a plot is always "too narrow" compared with a band for the whole $S(t)$.

Definition 6.7: Confidence band

$[L(t), U(t)]$ is a $100(1-\alpha)\%$ confidence band (CB) for $S(t)$ on $t_L \le t \le t_U$ if

$$P\{L(t) \le S(t) \le U(t) \text{ for all } t_L \le t \le t_U\} = 1-\alpha,$$

where $t_L \ge$ the smallest and $t_U \le$ the largest observed event time.

4.2 Equal probability (EP) bands

- Define

$$a_L = \frac{n\sigma_S^2(t_L)}{1+n\sigma_S^2(t_L)}, \qquad a_U = \frac{n\sigma_S^2(t_U)}{1+n\sigma_S^2(t_U)}, \qquad 0 < a_L \le a_U < 1.$$

- EP bands are **proportional to the pointwise confidence bounds**.
- From Table C.3 (K&M Appendix C) find the coefficient $c_\alpha(a_L, a_U)$ and build the band over $[t_L, t_U]$ by replacing $Z_{1-\alpha/2}$ in Definition 6.1 and Theorems 6.2–6.3 with $c_\alpha(a_L, a_U)$. (E.g. for a 90% band, $c_{0.10}(0.10, 0.40) = 2.4721$, larger than $Z_{0.95} = 1.645$.)

4.3 Hall–Wellner (HW) bands

- **Not** proportional to the pointwise bounds: wider for small t and narrower for large t.
- Find $k_\alpha(a_L, a_U)$ in Table C.4 and replace $Z_{1-\alpha/2}\sigma_S(t)$ in Definition 6.1 and Theorems 6.2–6.3 with

$$k_\alpha(a_L, a_U)\left[1+n\sigma_S^2(t)\right]n^{-1/2}.$$

4.4 Performance and sample-size advice

- The EP *linear* band performs poorly when $n < 200$: coverage is considerably below nominal.
- Log-log and arcsine-square-root EP bands have approximately correct coverage for $n \geq 20$; the arcsine EP band seems slightly better than the log-log EP band.
- For HW bands, all three versions perform reasonably well with as few as 20 observed events.

EP	Sample size	HW	Sample size
Linear	≥ 200	Linear	≥ 20
Log-log	≥ 20	Log-log	≥ 20
Arcsine	≥ 20	Arcsine	≥ 20

Example 6.8: Bone marrow transplant, ALL patients (K&M Table 4.3)

For 38 ALL patients, the KM estimate drops from $\hat{S}(1) = 0.9737$ ($Y = 38$) to $\hat{S}(662) = 0.3531$ ($Y = 13$), staying flat thereafter to the last (censored) time 2081; the Nelson–Aalen estimate reaches $\tilde{H}(662) = 1.0152$ with $\sigma_H = 0.2185$. Comparing 95% pointwise CIs, the EP band and the HW band for disease-free survival on roughly 100–600 days (K&M Fig. 4.5): both bands are wider than the pointwise intervals, and the HW band is wider than the EP band at early times but narrower later, as described above.

5 Selected exercises

Motivation. The interval formulas of this lecture are linked: the log interval for H is the log-log interval for S in disguise, the arcsine interval for H is the arcsine interval for S read through $H = -\log S$, and without censoring the band constants reduce to familiar empirical-process objects.

Exercise 6.1 (Log interval for H versus log-log interval for S)

Let $\tilde{H}(t) > 0$ and $\tilde{S}(t) = e^{-\tilde{H}(t)}$.
(a) Show that mapping the log-transformed CI for $H(t)$ through $S = e^{-H}$ gives $(\tilde{S}(t)^{\phi}, \tilde{S}(t)^{1/\phi})$, which is the log-log CI of Theorem 6.2 with $\hat{S}$ replaced by $\tilde{S}$ and σ_S by σ_H.
(b) Show that $\sigma_H^2(t) \leq \sigma_S^2(t)$ when all $Y_j > d_j$ ($t_j \leq t$), with

$$\sigma_S^2(t) - \sigma_H^2(t) = \sum_{t_j \leq t} \frac{d_j^2}{Y_j^2(Y_j - d_j)}.$$

Solution [Supplement].

(a)

Step 1. $H \mapsto e^{-H}$ is strictly decreasing, so the image of $[\tilde{H}/\phi, \phi\tilde{H}]$ is $[e^{-\phi\tilde{H}}, e^{-\tilde{H}/\phi}] = [\tilde{S}^{\phi}, \tilde{S}^{1/\phi}]$.

Step 2. With $\hat{S} \to \tilde{S}$ and $\sigma_S \to \sigma_H$, the constant of Theorem 6.2 is

$$\theta = \exp\left\{\frac{Z_{1-\alpha/2}\sigma_H}{\log \tilde{S}}\right\} = \exp\left\{-\frac{Z_{1-\alpha/2}\sigma_H}{\tilde{H}}\right\} = \frac{1}{\phi}.$$

Hence $(\tilde{S}^{1/\theta}, \tilde{S}^{\theta}) = (\tilde{S}^{\phi}, \tilde{S}^{1/\phi})$, the interval of Step 1. Both are Wald intervals on the scale $\log H = \log(-\log S)$.

(b) Termwise,

$$\frac{d_j}{Y_j(Y_j-d_j)}-\frac{d_j}{Y_j^2}=\frac{d_j\{Y_j-(Y_j-d_j)\}}{Y_j^2(Y_j-d_j)}=\frac{d_j^2}{Y_j^2(Y_j-d_j)}\geq 0;$$

sum over $t_j \leq t$. The relative difference of the j-th terms is $d_j/(Y_j-d_j)$, negligible unless the risk set is small, so the two intervals in (a) differ mainly in the tail. ■

Exercise 6.2 (Derivation of the arcsine-square-root interval for H)

Derive the arcsine-square-root CI for $H(t)$ stated in this lecture from Theorem 6.3.

Solution [Supplement].

Step 1 (standard error on the S scale). By the delta method for $\tilde{S}=e^{-\tilde{H}}$, $\mathrm{SE}(\tilde{S})\approx \tilde{S}\,\sigma_H$. So σ_H plays the role of $\sigma_S=\mathrm{SE}(\hat{S})/\hat{S}$, and Theorem 6.3 applies with $\hat{S}\to\tilde{S}$, $\sigma_S\to\sigma_H$.

Step 2 (rewrite in $\tilde{H}$). $\arcsin(\tilde{S}^{1/2})=\arcsin(e^{-\tilde{H}/2})$ and

$$\Big(\frac{\tilde{S}}{1-\tilde{S}}\Big)^{1/2}=\Big(\frac{e^{-\tilde{H}}}{1-e^{-\tilde{H}}}\Big)^{1/2}=(e^{\tilde{H}}-1)^{-1/2}.$$

So the angle interval is $[v_-,v_+]$ with

$$v_\pm=\mathrm{clip}_{[0,\pi/2]}\Big\{\arcsin\left(e^{-\tilde{H}/2}\right)\pm 0.5Z_{1-\alpha/2}\sigma_H(e^{\tilde{H}}-1)^{-1/2}\Big\},$$

and the CI for S is $[\sin^2 v_-,\sin^2 v_+]$.

Step 3 (back to H). $H=-\log S=-2\log\sin v$ is strictly decreasing in $v\in(0,\pi/2]$, so the endpoints swap: the CI for H is $[-2\log\sin v_+,\ -2\log\sin v_-]$, which is the stated formula (the $\min[\pi/2,\cdot]$ goes with the lower limit and the $\max[0,\cdot]$ with the upper). If $v_-=0$ the upper limit is $+\infty$. ■

Exercise 6.3 (Band constants without censoring)

Let $a(t)=n\sigma_S^2(t)/\{1+n\sigma_S^2(t)\}$, so that $a_L=a(t_L)$ and $a_U=a(t_U)$.
(a) Show that a is nondecreasing in t with values in $[0,1)$.
(b) Without censoring, show that $a(t)=1-\hat{S}(t)=\mathbb{F}_n(t)$ whenever $\hat{S}(t)>0$.
(c) Without censoring, show that the linear HW band is $\hat{S}(t)\pm k_\alpha(a_L,a_U)\,n^{-1/2}$, a band of constant width.

Solution [Supplement].

(a) $\sigma_S^2(t)$ is a sum of nonnegative terms over $t_j\leq t$, hence nondecreasing in t, and $x\mapsto x/(1+x)$ is increasing from $[0,\infty)$ onto $[0,1)$.

(b) By Exercise 5.1(b), $\sigma_S^2(t)=\{1-\hat{S}(t)\}/\{n\hat{S}(t)\}$, so $1+n\sigma_S^2=1/\hat{S}$ and

$$a(t)=\frac{(1-\hat{S})/\hat{S}}{1/\hat{S}}=1-\hat{S}(t)=\mathbb{F}_n(t)$$

by Exercise 5.1(a). So $[a_L,a_U]$ is the range of the empirical cdf over $[t_L,t_U]$, the natural time scale for the Brownian-bridge limit of $\sqrt{n}(\mathbb{F}_n-F)$.

(c) The HW band replaces $Z_{1-\alpha/2}\sigma_S(t)$ in Definition 6.1 by $k_\alpha[1+n\sigma_S^2(t)]n^{-1/2}=k_\alpha n^{-1/2}/\hat{S}(t)$ (by (b)). The half-width $\hat{S}(t)\cdot k_\alpha n^{-1/2}/\hat{S}(t)=k_\alpha n^{-1/2}$ does not depend on t: a Kolmogorov–Smirnov-type band. ■